Deep Ensembles for Predictive Uncertainty

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Abstract

State the task, datasets, compared methods, and the main empirical conclusions in 100–150 words. Write this last.

1 Question and experimental design

Explain why predictive uncertainty matters. State the task, 2-D synthetic sanity check, realistic high-dimensional dataset (recommended: Fashion-MNIST), train/test splits, preprocessing, and fixed seeds. Define train/test accuracy or error, predictive entropy, expected data uncertainty, and mutual information. State how hyperparameters were selected without using the test set.

2 Methods

2.1 Standard deep ensembles

Describe independently initialized models trained on identical data with distinct data-order seeds. Give the averaged predictive distribution $p(y \mid x, \mathcal{D}) \approx M^{-1} \sum_m p_m(y \mid x)$. State architecture, optimizer, learning rate, epochs, and ensemble sizes. Explain the L2 loss term and its isotropic zero-mean Gaussian-prior interpretation; report its coefficient.

2.2 Comparison method: MC dropout

When implemented, describe dropout placement/rate, number of stochastic passes, training objective, and retaining dropout at test time. Explain approximate Bayesian sampling and keep setup comparable to the ensemble.

3 Results on the 2-D synthetic task

Show decision probabilities and uncertainty surfaces. Verify uncertainty behavior away from observations and at ambiguous boundaries; discuss failures. Compare train/test performance. Vary $M \in \{1, 2, 5, 10\}$ and interpret accuracy/error, predictive entropy, and mutual information.

4 Realistic high-dimensional dataset

Report the same train/test metrics and uncertainty summaries on Fashion-MNIST (or justify a replacement). Include a table comparing standard deep ensembles and MC dropout at matched compute and an ensemble-size plot. Analyze uncertain correct and incorrect examples and whether uncertainty identifies likely mistakes. Do not claim calibration from a single mean uncertainty number.

5 Comparison and discussion

Directly compare methods on accuracy/error, uncertainty behavior, compute/memory, and sensitivity to ensemble size or MC sample count. Explain limitations: finite seeds, dataset scope, hyperparameter sensitivity, and the difference between in-distribution ambiguity and out-of-distribution uncertainty. Mention unexpected results and plausible explanations.

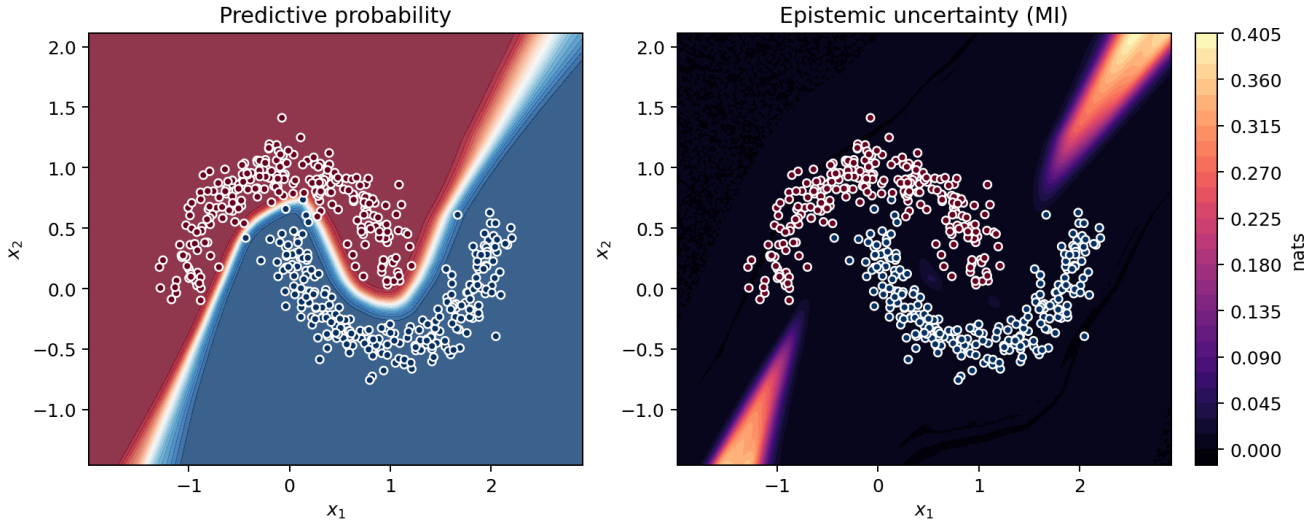


Figure 1: Standard deep-ensemble predictive probability and epistemic uncertainty on two moons. Explain observed regions.

6 Conclusion

Give two or three concise takeaways about which method is more useful for each goal, backed by the results.

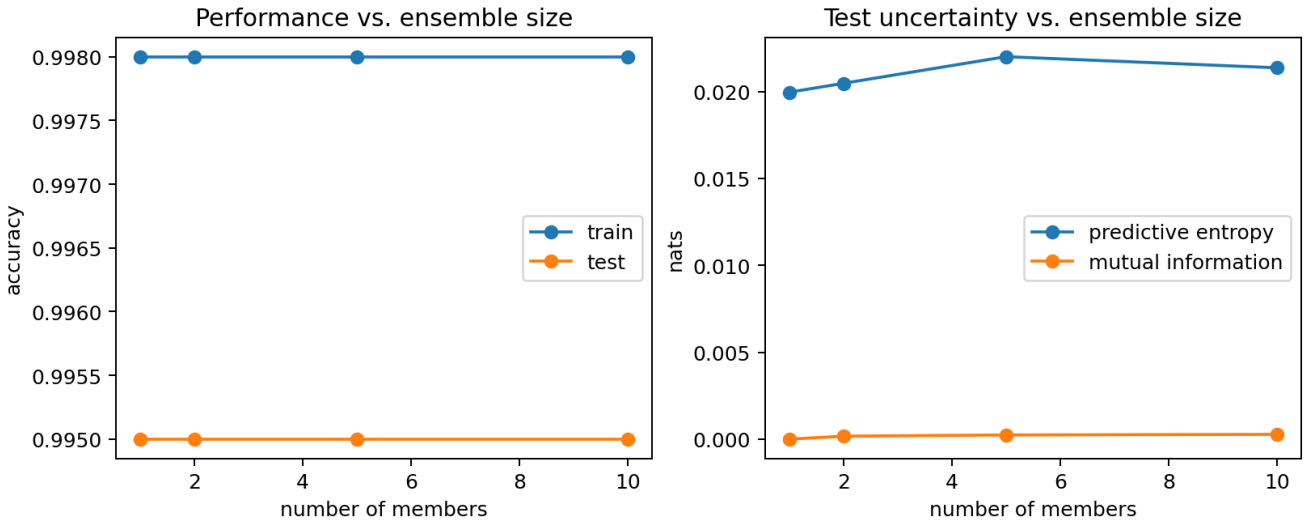


Figure 2: Training and test performance plus uncertainty summaries as ensemble size changes. Discuss diminishing returns and variability.